

Algebra II with Trigonometry

Chapter 5.3

Study guide

Saved tutoring response with MathJax rendering. Original PDF file:
[4401269_alg-2trig-h-chp-5-3-note-docx.pdf](#)

Certainly! Here's a detailed study guide based on the provided document. Each request is addressed in a clear, organized manner:

1. Short Summary of the Document

This document is a lesson from Algebra 2/Trigonometry Honors covering **trigonometric functions of angles**. It explains the definition and evaluation of trig functions for any angle, summarizes fundamental trig identities, and demonstrates the application of trig concepts to triangle areas and real-world word problems. The document provides tables, definitions, example problems, and diagrams for visual aid.

2. Core Definitions, Theorems, and Formulas

A. Definitions

- **Standard Position:** An angle with its vertex at the origin and initial side along the positive x-axis.
- **Reference Angle:** The acute angle formed by the terminal side of θ and the x-axis.
- **Quadrantal Angle:** An angle whose terminal side lies on the x- or y-axis.

B. Trigonometric Functions (using point $P(x, y)$, distance

$$r = \sqrt{x^2 + y^2}$$

- $\sin \theta = \frac{y}{r}$

- $\cos \theta = \frac{x}{r}$
- $\tan \theta = \frac{y}{x} \ (x \neq 0)$
- $\csc \theta = \frac{r}{y} \ (y \neq 0)$
- $\sec \theta = \frac{r}{x} \ (x \neq 0)$
- $\cot \theta = \frac{x}{y} \ (y \neq 0)$

C. Signs of Trigonometric Functions by Quadrant

Quadrant	Positive Functions	Negative Functions
I	all	none
II	sin, csc	cos, sec, tan, cot
III	tan, cot	sin, csc, cos, sec
IV	cos, sec	sin, csc, tan, cot

D. Trigonometric Identities

Reciprocal Identities

- $\csc \theta = \frac{1}{\sin \theta}$
- $\sec \theta = \frac{1}{\cos \theta}$
- $\cot \theta = \frac{1}{\tan \theta}$
- $\tan \theta = \frac{\sin \theta}{\cos \theta}$
- $\cot \theta = \frac{\cos \theta}{\sin \theta}$

Pythagorean Identities

- $\sin^2 \theta + \cos^2 \theta = 1$
- $\tan^2 \theta + 1 = \sec^2 \theta$
- $1 + \cot^2 \theta = \csc^2 \theta$

E. Triangle Area Using Trig

- For triangle with sides a , b and included angle θ :

$$A = \frac{1}{2}ab \sin \theta$$

3. Worked Study Guide

A. Finding Trig Functions at Any Angle

1. **Find the reference angle:** Reduce θ to an angle between 0° and 90° with the x-axis. 2. **Determine the sign:** Use the quadrant to find the sign. 3. **Calculate value:** Use reference angle and sign.

B. Example:

Find $\csc 135^\circ$:

- Reference angle $= 180^\circ - 135^\circ = 45^\circ$
 - 135° is in Quadrant II, sine is positive.
 - $\sin 45^\circ = \frac{\sqrt{2}}{2}$, so $\csc 45^\circ = \frac{1}{\frac{\sqrt{2}}{2}} = \sqrt{2}$
 - Answer: $\csc 135^\circ = \sqrt{2}$
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C. Applying Trigonometric Identities

If $\sin \theta = \frac{3}{5}$, θ in Quadrant II:

- $\cos \theta = -\sqrt{1 - (\sin \theta)^2} = -\sqrt{1 - \frac{9}{25}} = -\sqrt{\frac{16}{25}} = -\frac{4}{5}$

D. Area of Triangle Example

Given $a = 10$, $b = 22$, $\theta = 10^\circ$:

- $A = \frac{1}{2}ab \sin \theta \approx \frac{1}{2}(10)(22)\sin 10^\circ \approx 110 \sin 10^\circ \approx 110 \cdot 0.1736 \approx 19.10$

E. Word Problem Application

Height of Rocket:

- Distance from observer $= 1 \text{ mile} = 5280 \text{ feet}$.
 - $h = 5280 \tan \theta$, where θ is the angle of elevation.
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4. Practice Problems (with Answers)

Practice 1

Find the reference angle for -199°

- 199° is 19° past 180° , so reference = $199^\circ - 180^\circ = 19^\circ$.
- Since it's negative, the reference angle is 19° past -180° , so reference = 19° .

Practice 2

Evaluate $\tan\left(-\frac{4\pi}{3}\right)$

- $-\frac{4\pi}{3} + 2\pi = \frac{2\pi}{3}$, reference angle = $\frac{\pi}{3}$.
- $\tan\left(-\frac{4\pi}{3}\right) = \tan\left(\frac{2\pi}{3}\right) = -\sqrt{3}$

Practice 3

Which quadrant if $\tan \theta < 0$ and $\sin \theta < 0$?

- Tan negative: II or IV. Sin negative: III or IV. So Quadrant IV.

Practice 4

Find area: triangle sides 8, 13, included angle 42° .

- $A = \frac{1}{2} \cdot 8 \cdot 13 \cdot \sin 42^\circ \approx 52 \cdot \sin 42^\circ \approx 52 \cdot 0.6691 \approx 34.79$

Practice 5

If $\sin \theta = \frac{3}{10}$, $\sec \theta > 0$, find $\cos \theta$

- $\cos \theta = \sqrt{1 - \left(\frac{3}{10}\right)^2} = \sqrt{1 - \frac{9}{100}} = \sqrt{\frac{91}{100}} = \frac{\sqrt{91}}{10}$
- Since $\sec \theta > 0$, $\cos \theta > 0$.

5. Assumptions and Ambiguities

Assumptions:

- All angle measurements are in degrees unless otherwise specified or in radians when π is shown.

- Reference angle method applies consistently to both positive and negative angles.
- Some problems refer to included images/graphs. Descriptions are given in place of image details when not visible.
- Calculations rounded as necessary, with sin/cos values often approximated to 4 decimal places unless exact values are known.

Ambiguities:

- Some solution paths depend on quadrant placement, always clarified in the solution.
- Not all notation in the PDF is clearly typeset; standard mathematical interpretation is used where needed.

If you have a particular problem or example from the text you want a detailed solution to, just ask!